TEXT MINING WITH MACHINE LEARNING

Principles and Techniques



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Text Mining with Machine Learning Principles and Techniques

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Preface

The aim of this book is to make it easier for those interested in text mining - one of the branches of modern information technology - to enter an interesting area focused on analyzing texts in natural languages. Machine learning has been chosen as an analytical tool because it is a promising organic part of computer science and artificial intelligence that supports successful exploration of big volumes of text data using advanced classification and clustering methods. Machine learning now has available a large number of software tools, both commercial and free, so there is no need to implement them and a researcher can focus directly on their use for given data and applications.

The authors of this book have chosen demonstrations of certain basic as well as popular implementations of machine learning algorithms in an advanced interpreted programming language R, which is freely available for a variety of operating systems and provides an appropriate communication interface over RStudio without the need for complex learning. In no case is this book primarily focused on advanced computer scientists, programmers, or mathematicians, although basic programming skills and higher mathematical knowledge (linear algebra, statistics, function analysis) are required to understand algorithms, use them and, of course, interpret their results. For interested readers, chapters have references to literature which can suitably complement the topics described. Text mining has already come to an advanced/a well-developed form, driven by the need to explore texts as specific data type, in particular large text data files on the Internet from social networks, product or service reviews, e-mails, blogs, and many forums. Perhaps every e-mail user today is aware of the facility to filter messages (for example, spam removal or topic grouping), which is just one of many possible applications of the text mining industry.

The book covers the introduction to text mining by machine learning, introduction to the R programming language, structured text representation, classification and easy introduction to the most common classification algorithms (Bayes classifier, nearest neighbors, decision trees, random forest, sup-

port vector machines, and deep learning), followed by clustering. The final two chapters deal with word embedding and feature selection problem areas. The authors of this book would like to express their hope that the book will find a number of interested readers and will be used to familiarize them with the attractive sector of artificial intelligence and machine learning applied to the difficult area of processing data produced by humans and not by machines.

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Authors' Biographies

Jan Žižka is currently involved in consultations in the field of machine learning and data mining. For many years he worked as a system programmer, developer of advanced software systems, and researcher. For the last 25 years, he has devoted himself to artificial intelligence and machine learning, especially text mining. He has also worked at a number of universities in the Czech Republic and abroad, including research institutes. He is the author and co-author of approximately 100 international publications.

František Dařena works as an associate professor, the guarantor of the doctoral degree program System Engineering and Informatics, and head of the Text Mining and Natural Language Processing group at the Department of Informatics, Faculty of Business and Economics, Mendel University Brno. He is the author of several publications in international scientific journals, conference proceedings, and monographs; member of the editorial boards of international journals, and editor-in-chief of International Journal in Foundations of Computer Science & Technology. His research areas include text/data mining, intelligent data processing, and machine learning.

Arnošt Svoboda is an expert in programming and his area includes programming languages and systems such as R, Assembler, Matlab, PL/1, Cobol, Fortran, Pascal, and others. He started his career as a system programmer. For the last 20 years, Arnos˘t has worked as a teacher and researcher at Masaryk University in Brno (Czech Republic), Faculty of Economics and Administration, Department of Applied Mathematics and Computer Science. At present, he is interested in machine learning and data mining



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Chapter 1

Introduction to Text Mining with Machine Learning

1.1 Introduction